Interactive Sinhala Text Reader for the Visually Impaired with the assistance of OCR, NLP, and AI.

Group R26-DS-002 | Sri Lankan Institute of Information Technology

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The Problem

Accessibility Gap in Sri Lanka

- Approximately 2.2 billion people globally suffer from vision impairment, with 1.1 million affected in Sri Lanka alone.
- Existing assistive tools like JAWS and Google Lookout are English-centric and offer no functional Sinhala script support.
- Sinhala OCR systems fail on stylized headline fonts, which carry the highest semantic density in Sinhala documents and standard technologies tend to produce errors.



Visually impaired Sinhala speakers are excluded from printed information due to the absence of affordable, script-aware assistive technology.

Evidence and Credibility

Visual Impairment Scale

- 2.2 billion people globally live with some form of vision impairment (WHO, 2026).
- 1.1 million individuals in Sri Lanka are/might be affected, survey shows 40+ adults have 32.3% prevalence, combining mild, moderate, severe, and blind,from~6.8 million at that time (MOH-Visual Impairment Survey,14/15)

Sinhala OCR Failure Rate

- Surya-OCR achieves only 19.57% WER on clean Sinhala text; assuming performance is significantly worse on complicated layouts.
- No major OCR engines like Tesseract or Surya-OCR have demonstrated robust accuracy on stylized Sinhala headline fonts, per recent Sinhala OCR benchmarks

Tool Accessibility Gap

- High-end wearables like OrCam exceed USD 2,000 and have no Sinhala language support.
- All major screen readers (JAWS, NVDA, Google Lookout) are limited to Latin-script text only.

Research Ecosystem Gap

- Survey of Sinhala NLP tools confirms limited systematic work addresses display or headline typography (De Silva, 2019).
- Post-OCR correction remains rule-based and word-level; no context-aware transformer model exists for Sinhala.

These figures confirm an urgent, unmet need: no affordable, Sinhala-capable, font-invariant assistive reading solution currently exists.

Users, Stakeholders and Impact

Users and Stakeholders

- Primary users are visually impaired and elderly Sinhala-speaking individuals who rely on newspapers, textbooks and other documents for daily information.
- Secondary users include OCR developers and research institutions seeking a font-invariant Sinhala recognition, context aware Error correction, conversational Sinhala voice modules for integration.
- Tertiary stakeholders include Sri Lankan disability advocacy groups, government education agencies, and the open-source NLP community.

Impact on Users and Society

- Enables affordable, hands-free access to Sinhala printed information for visually impaired Sri Lankans.
- Directly advances UN SDG 10 (Reduced Inequalities) and SDG 4 (Quality Education) by removing information barriers for persons with disabilities.
- Open-source release will benefit the Sinhala NLP research ecosystem and enable community-driven improvement of assistive tools.

Existing Solutions and Their Limitations

Tesseract / Surya-OCR	• Appearance-based models degrade significantly on stylized Sinhala headline fonts due to training distribution mismatch. • No evaluated engine addresses the font-invariant recognition gap in Sinhala display typography.
Wearable Devices (OrCam, Envision Glasses)	• Priced above USD 2,000, making them inaccessible to the majority of Sri Lankan users. • No Sinhala language support; limited to standard printed body text in Latin-script environments.
Rule-Based Post-OCR Correction (SinSpell, Weerasinghe)	• Word-by-word correction with no sentence-level context results in poor generalization to unseen patterns. • Optimized for human typing errors, not for visual noise artifacts introduced by OCR scanning.

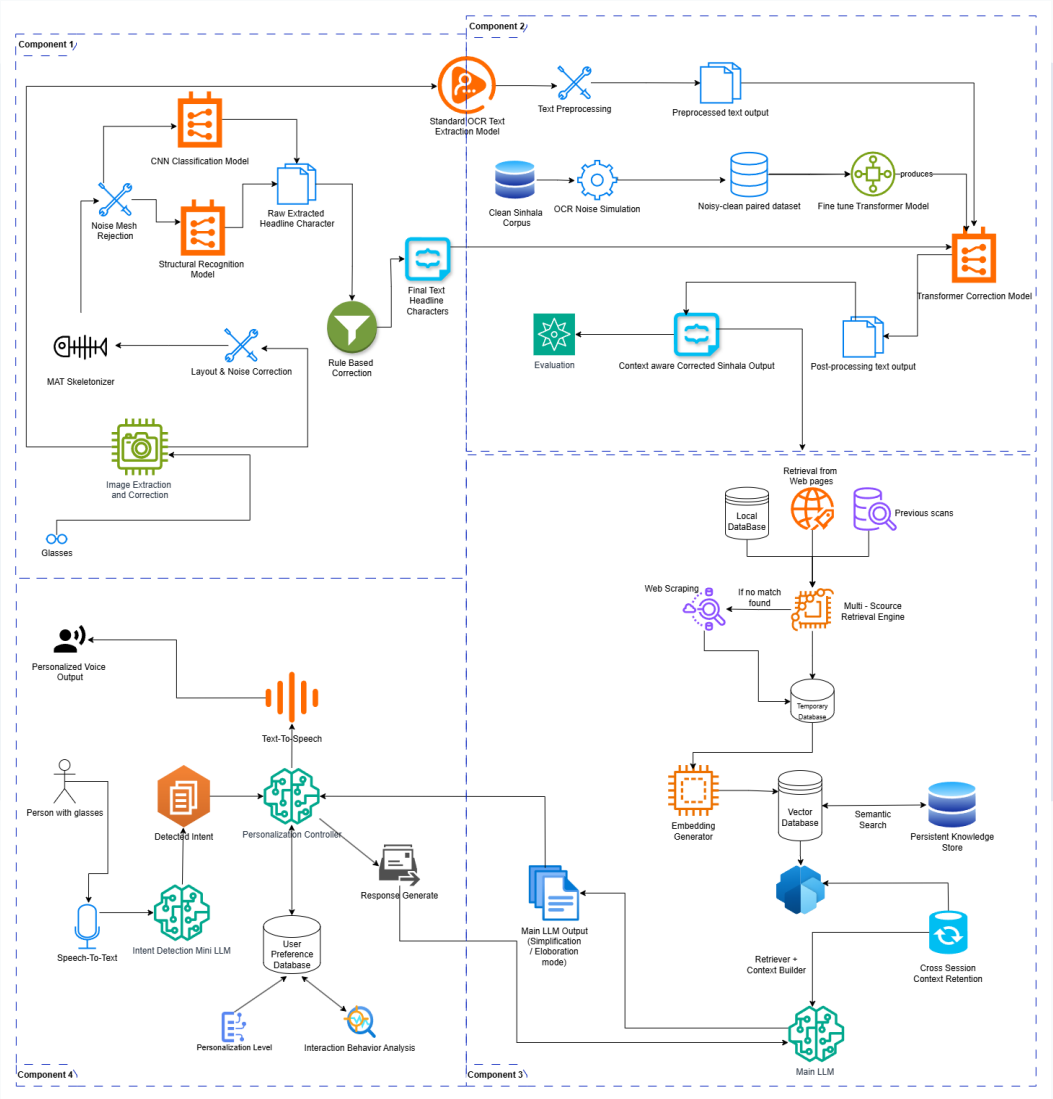
No existing solution addresses font-invariant Sinhala headline recognition with context-aware error correction in a single affordable pipeline.

Proposed Solution

Solution Overview

- An application that combines topological Sinhala OCR (Component 1) with transformer-based post-OCR error correction (Component 2).
- Retrieves contextual knowledge via RAG and generates simplified or elaborated responses using a Main LLM (Component 3).
- Personalizes and delivers the final output as natural Sinhala voice, enabling bidirectional voice interaction from the user (Component 4) .

Key differentiator: the only known end-to-end pipeline combining font-invariant Sinhala headline OCR with context-aware transformer correction along with and personalized generation in a fully offline, wearable form factor.



Individual Students' Components

M.J.J Muthunayaka (IT22072238)

Topological Sinhala Character Recognition via Medial Axis Transform

- Applies MAT-based skeletonization to reduce Sinhala characters to font-invariant structural representations for recognition.
- Classifies headline characters using a CNN-LSTM model and an exploratory skeleton route matching approach.

H.P.I Madhusanka (IT22259134)

Context-Aware Sinhala Post-OCR Error Correction Model

- Detects OCR-corrupted tokens using a fine-tuned SinBERT binary classifier operating over full sentence context.
- Corrects flagged tokens using a fine-tuned mT5 sequence-to-sequence model to produce reliable Sinhala text output.

W.G.N Devindi (IT22120670)W.G.

Advanced Document Retrieval and Information Generation

- Retrieves relevant content from local databases, web sources, and previous scans using vector embeddings and semantic search.
- Generates simplified or elaborated responses via a RAG-powered Main LLM with cross-session context retention.

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Adaptive Conversational Personalization and Voice Interaction

- Transcribes user speech, detects intent via a Mini LLM, and applies a personalization layer from stored user preferences.
- Delivers the final tailored response as natural Sinhala voice output via Text-to-Speech synthesis.

IT22072238 | MUTHUNAYAKA M.J.J



Topological Sinhala Character Recognition

via Medial Axis Transform

Problem Definition and Novelty

The Problem

- Existing OCR tools fail on stylized Sinhala headline fonts where the most information-rich text in documents.
- Appearance-based models degrade when font style deviates from training data, and there are so many fonts which regular changes and deviate.
- No published system specifically targets font-invariant Sinhala display typography.



NOVELTY — First application of MAT-based topological skeletonization as a primary recognition mechanism for stylized Sinhala display typography.

Research Objectives

Main Objective

Design and evaluate a MAT-based pipeline that accurately recognizes stylized Sinhala headline text, improving CER and WER over standard OCR baselines.

01 Develop a MAT skeletonization pipeline preserving single pixel-wide character structure, reducing stroke thickness sensitivity, Bleeding & Fragmentation.

02 Explore Skeleton Route Matching as a topological alternative approach.

03 After conversion, isolate headline text using various thresholding methods and Connected Component Analysis while retaining diacritics.

04 Benchmark the pipeline against Tesseract using CER,WER, and processing speed.





Research Methodology and Technologies

1. **Image Extraction & Correction**

   Skew and brightness correction on camera-captured input.

2. **Layout & Noise Isolation**

   Thresholding + CCA to anchor bodies and retain diacritics, including binarization.

3. **MAT Skeletonization**

   Reduces characters to single-pixel skeletons; extracts endpoints and junctions.

4. **Noise Mesh Rejection**

   Tortuosity filter to removes unnecessary paths and retain valid strokes.

5. **Character Recognition**

   Start with basic classification, switch to a Skeleton Route Matching as alternative.

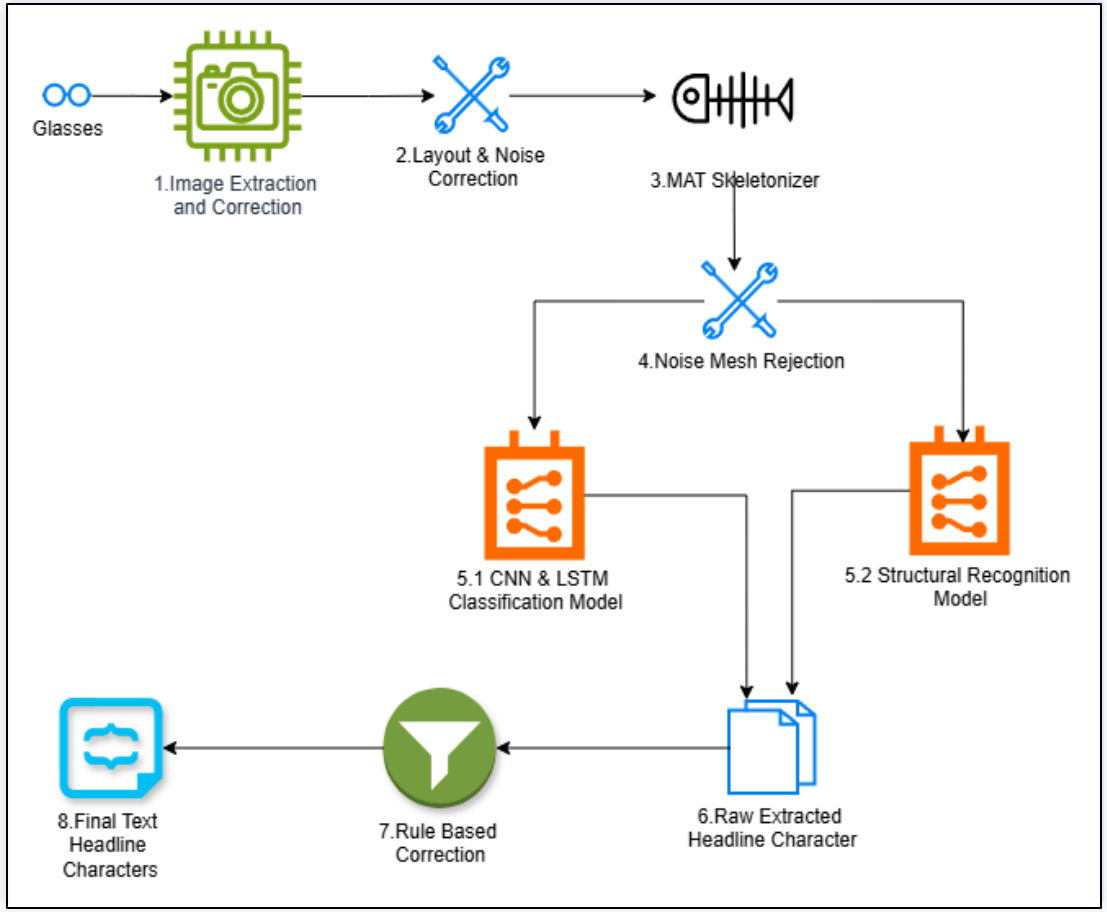
TOOLS & TECHNOLOGIES

- **Python**
  Core language
- **OpenCV**
  Preprocessing & thresholding
- **Scikit-Image**
  MAT & skeletonization
- **NumPy / SciPy**
  Numerical & graph operations
- **TensorFlow / PyTorch**
  CNN-LSTM model
- **Tesseract / Surya OCR**
  Baseline comparison

System Overview

Stages of the Component

- Image capture and correction
- Layout isolation via CCA and thresholding
- MAT skeletonization of characters
- Tortuosity-based noise rejection
- Classification + rule-based correction
- Final headline text output to Component 2



Component 1 runs in parallel with Component 2, feeding headline text as context for body-text correction.

Dependencies · Validation · Evaluation · Data

DEPENDENCIES

- Output feeds directly into Component 2 as headline context.
- All tools are open-source : Python, OpenCV, Scikit-Image, TensorFlow.
- Requires an auto-focus industrial webcam for best input quality but for the research phase, a high-quality imaging source mounted at a fixed perspective

VALIDATION & EVALUATION

- Metrics: Character Error Rate, Word Error Rate, and processing speed will be used to measuring.
- Benchmarked against Tesseract and Surya-OCR on stylized headline images.
- Aiming for a higher accuracy or higher processing speed against industry standard tools for final evaluation.

DATA AVAILABILITY

- Digital print media from e-newspapers and government textbook PDFs.
- Manually captured document photos at realistic eye-level angles.
- Dataset made from publicly available Sinhala .TTF font files.

ETHICS & SECURITY

- Only publicly available, openly licensed content used — no personal data.
- All processing is local; no visual data transmitted to external servers.
- Basic undergraduate ethics guidelines for researches.

Summary

Problem	Standard OCR fails on stylized Sinhala headline fonts — no existing tool addresses this.
Approach	MAT skeletonization produces font-invariant 1-pixel structural representations.
Pipeline	Image correction → noise isolation → skeletonization → filtering → recognition model → Results
Evaluation	Benchmarked against Tesseract and Surya-OCR using CER, WER, and speed.
Impact	Open-source release for visually impaired Sinhala readers — advancing SDG 4 and SDG 10.

IT22259134 Madhusanka H.P.I



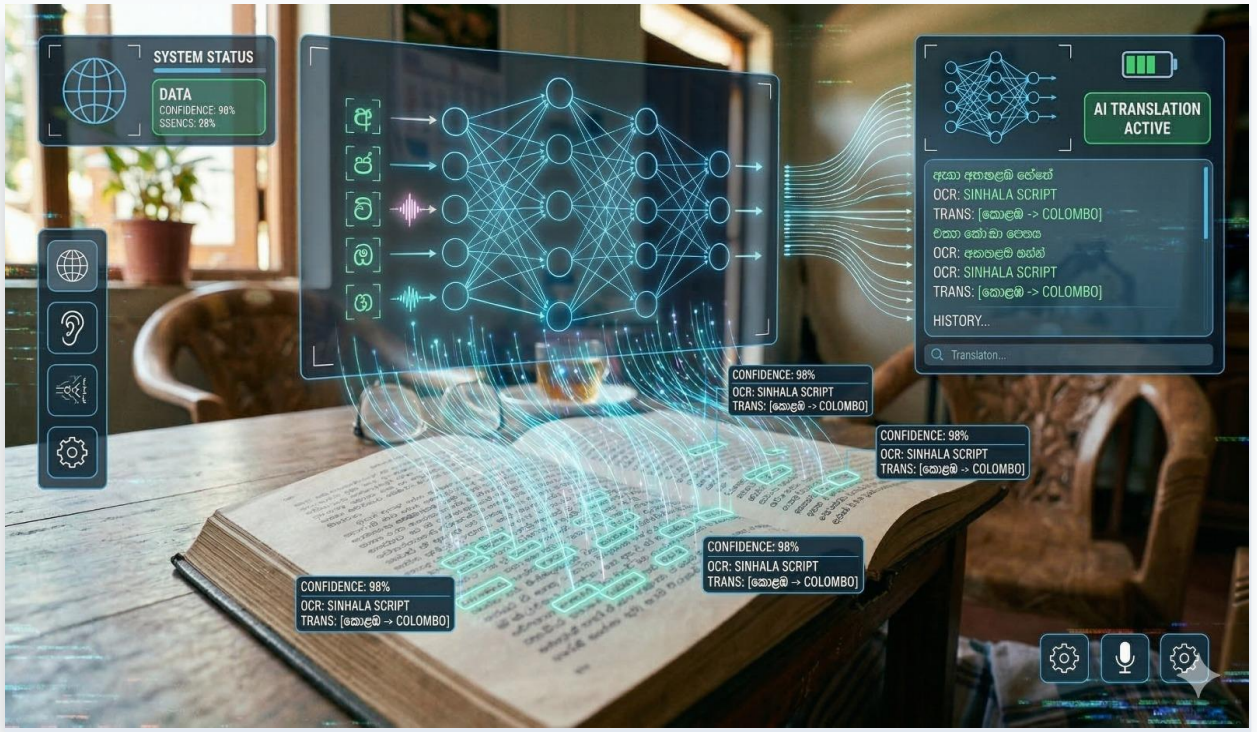
Context-Aware Sinhala Post OCR Error Correction Model

for Assistive Document Reading

Problem Definition & Novelty

The Problem

- ~150,000 blind Sri Lankans cannot read Sinhala newspapers or textbooks.
- Sinhala OCR makes around 19.57% word errors.
- Wrong OCR words are read aloud to the user → system becomes unreliable
- Existing Sinhala tools are rule-based, word-level, designed for typing errors.



Novelty: First Sinhala post-OCR correction using transformer model, First Sinhala OCR error dataset (~120,000 sentence pairs), Sinhala-specific OCR error taxonomy (5 error types)

Research Objectives

Main Objective : Develop a two-stage context-aware transformer based post-OCR correction system (SinBERT + mT5) that reduces CER and WER for Sinhala assistive reading

O1	Define a Sinhala-specific OCR error classification with five error categories.
O2	Construct a Sinhala OCR noise dataset of about 120,000 noisy–clean sentence pairs.
O3	Fine-tune SinBERT-Large as an OCR error token detector.
O4	Fine-tune mT5-small as a context-aware sequence corrector.
O5	Evaluate the pipeline against three baselines (raw OCR, rule-based, single-stage mT5) using CER, WER and over-correction.
O6	Confirm CPU inference time is suitable for real-time use.

Research Methodology & Technologies

Pipeline Stages

1. **Data Collection -** Scan Lankadeepa/Mawbima, run OCR, align with ground-truth text
2. **Noise Synthesis -** Inject synthetic OCR noise into clean Sinhala text based on real error patterns.
3. **SinBERT Fine-tuning -** Train token level error detector on noisy-clean pairs.
4. **mT5 Fine-tuning -** Train seq-to-seq corrector on sentences with marked errors.
5. **Evaluation -** Compare with three baselines using CER, WER, over-correction and speed.

TOOLS & TECHNOLOGIES

- **Python/ PyTorch**
  Training pipeline & experimentation
- **HuggingFace Transformers**
  Model fine-tuning & inference
- **Google Colab Pro**
  GPU compute for fine-tuning
- **BeautifulSoup**
  Sinhala news corpus scraping
- **Unicodedata NFC**
  Unicode normalization pre-processing
- **Tesseract / Surya OCR**
  Baseline comparison

Dependencies, Validation, Data & Ethics

System Dependencies

- Component 1 (OCR Engine) provides raw Sinhala text input via UTF-8.
- Component 3 (RAG) consumes corrected text and confidence JSON.
- Components 3 and 4 depend on Component 2 for clean text.

Validation & Evaluation

- Compare with three baselines: raw OCR, rule-based, single-stage mT5.
- Use metrics: CER, WER, over-correction rate, CPU latency.
- Test set uses only real OCR noise for honest performance.

Data Availability

- First Sinhala OCR correction dataset (~120,000 pairs).
- 30% real pairs from scanned Lankadeepa and Mawbima pages.
- 70% synthetic pairs from clean text with injected OCR-likenoise.

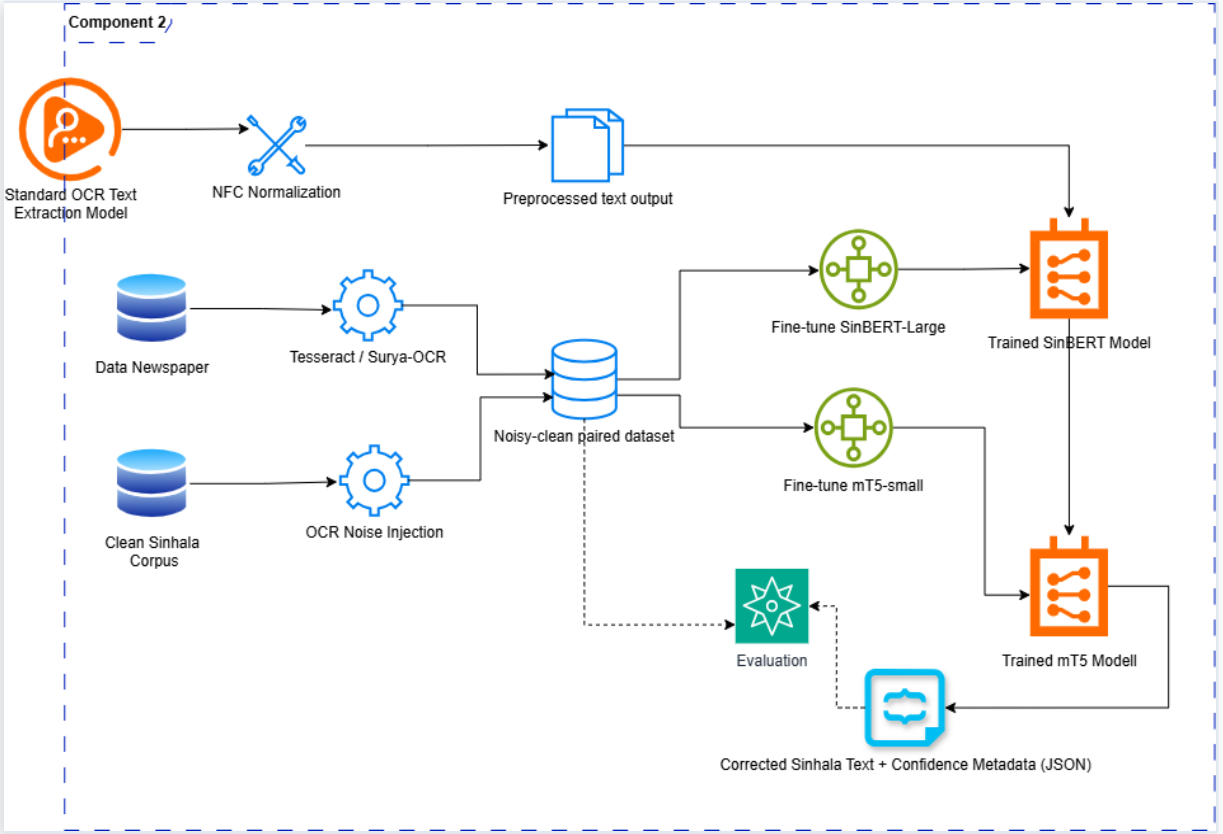
Ethics & Security

- All data from public newspapers and government websites.
- No personal data stored; processing fully on user device.

System Overview – Two-Stage Pipeline

Pipeline Stages

- Raw OCR Text → Unicode NFC Normalization
- SinBERT token classifier → labels tokens as CORRECT / ERROR with confidence
- Confidence threshold filter → low-confidence tokens kept as CORRECT
- mT5-small corrector → fixes flagged tokens using sentence context
- Corrected text + confidence metadata (JSON) → Component 3 → Component 4 reads to user



Summary

Problem	Sinhala OCR errors degrade assistive reading quality for blind and low-vision users
Approach	Two-stage pipeline: SinBERT detects error tokens. mT5 produces context-aware corrections
Pipeline	NFC normalization → SinBERT classification → confidence filter → mT5 correction → JSON output
Evaluation	CER, WER, over-correction rate and CPU latency vs three baselines on real OCR data
Impact	First Sinhala transformer-based OCR correction and dataset, enabling more reliable real-time assistive reading

IT22120670 | W.G.N DEVINDI



Advanced Document Retrieval and Information Generation

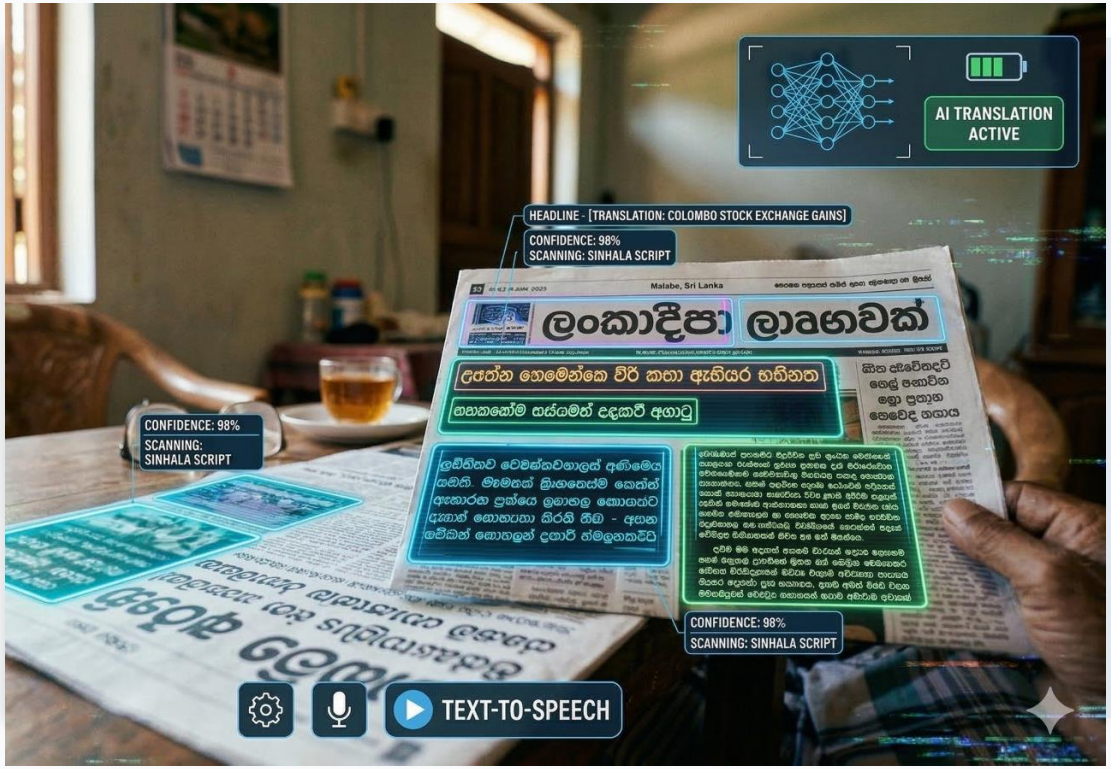
RAG Pipeline for Sinhala Assistive Reading — Component 3

R26-DS-002

Problem Definition & Novelty

The Gap in Sinhala Assistive Technology

- No Language tool exists to measure whether a Sinhala sentence is simple or complex, meaning current assistive systems read the same content to every user
- When OCR scans a Sinhala newspaper, character-level errors in conjuncts and diacritics cause standard retrieval systems to return the wrong article entirely
- Existing Sinhala reading systems produce only fixed, linear audio output with no way to ask questions, request a summary, or receive content adapted in real time.



NOVELTY — First Sinhala complexity-adaptive reading system that adjusts content difficulty to match each individual user.

Research Objectives

MAIN OBJECTIVE — Develop a system that reads Sinhala newspaper articles and automatically adjusts how simple or complex the content sounds based on who is listening..

O1 Retrieve the correct Sinhala newspaper article even when the OCR scan contains character errors.

O2 Build a knowledge base that automatically saves and updates itself with new Sinhala news articles without manual work.

O3 Create the first tool that measures how difficult a Sinhala paragraph is to read and classify it into three levels.

O4 Rewrite retrieved Sinhala content in simpler or more detailed language based on what the user requests.

Research Methodology & Technologies

1. **Data Collection**
   scraped from Lankadeepa and Mawbima using BeautifulSoup. - annotated for complexity across 3 difficulty levels

2. **Data Preprocessing**
   Unicode normalization and sentence tokenization using sinling -

3. **Two-Stage Document Retrieval**
   LaBSE embeds the article headline for coarse candidate search.

4. **RAG Pipeline & Knowledge Base**
   Retrieved content stored in ChromaDB local vector store with three-tier reliability scoring.

5. **Dynamic Text Adaptation**
   Instruct the LLM to simplify or elaborate

Tools & Technologies

Python
Core programming language

sinling
Sinhala tokenization & normalization

ChromaDB
Local Persistent vector store

LangChain
RAG chains orchestration

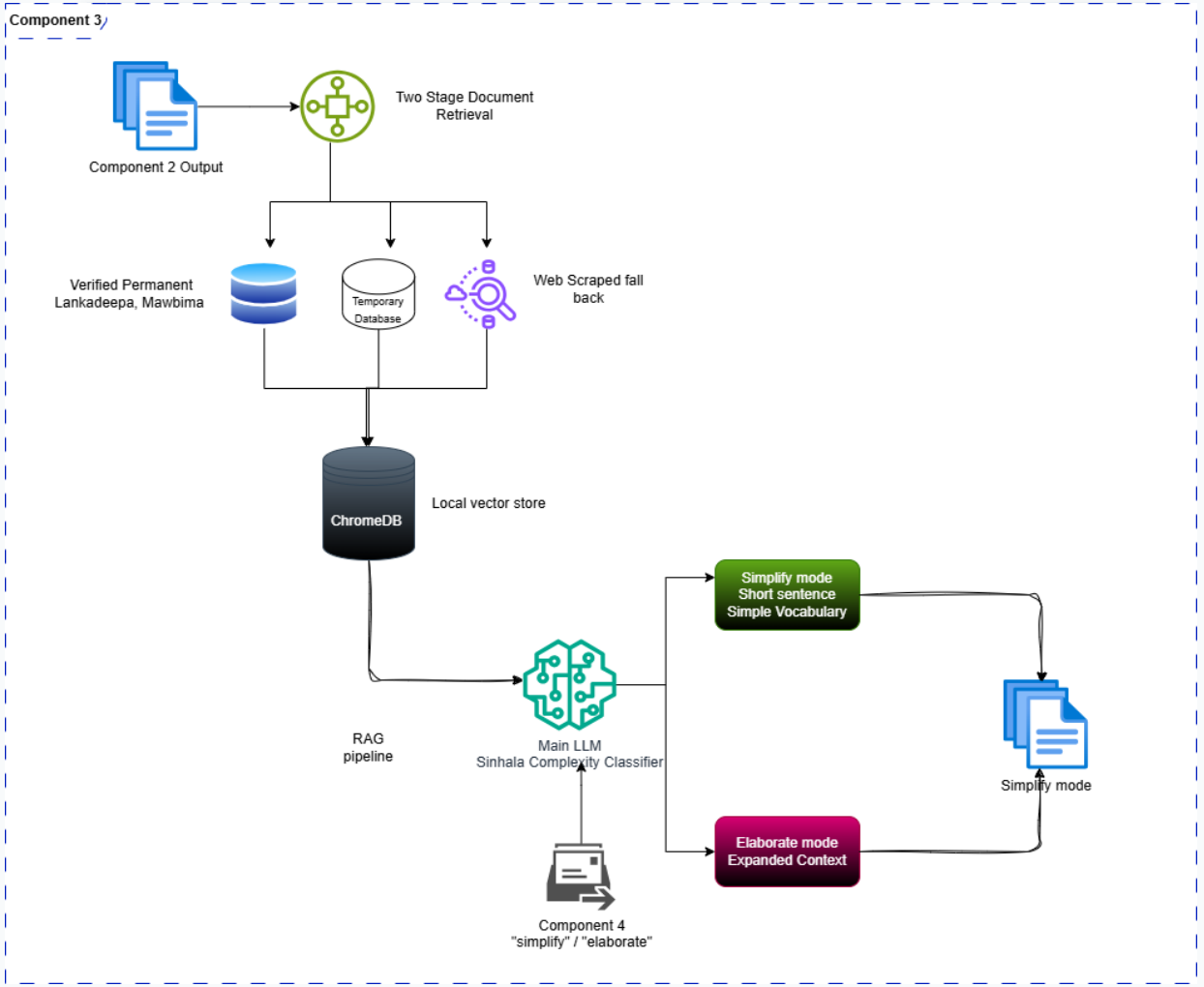
LaBSE
Multilingual headline embedding

Gemma 2B
Local LLM via Ollma

System Overview

Stages of the Component

- Sinhala document JSON received from Component 2
- Headline embedding and two-stage candidate retrieval
- Tiered knowledge base lookup and web scraping fallback
- RAG pipeline generation using retrieved context
- Complexity classification of retrieved content
- JSON response with metadata passed to Component 4 .



Dependencies, Validation, Data & Ethics

Dependencies

- Component 2 provides corrected Sinhala text as a structured JSON object
- Component 4 sends a real-time intent signal — "simplify" or "elaborate" — which triggers the adaptation mode in Component 3.

Validation & Evaluation

- Compare two-stage retrieval against single-stage baseline measure Top-1 and Top-3 precision

  End-to-end latency profiled using Python timeit under controlled single and multi-query loads.
- User acceptance testing with 5 visually impaired participants measuring task completion rates.

Data Availability

- Sinhala news articles scraped from Lankadeepa and Mawbima using BeautifulSoup — forms Tier 1 verified knowledge base.
- Paragraphs annotated for complexity across three levels by native Sinhala speakers via Google Forms.
- Textbook PDFs sourced from Educational Publications Department (edupub.gov.lk) for domain coverage.

Ethics & Security

- All scraped content sourced from publicly available Sinhala news portals and government websites — no copyrighted private data.
- Informed consent obtained from user study participants — no personal data retained after study completion.

Summary

Problem	400,000 visually impaired Sri Lankans have no affordable Sinhala assistive reading tool. Existing systems cannot adjust content difficulty for different users.
Approach	A complexity-adaptive RAG pipeline combining LaBSE, ChromaDB , and dynamic text adaptation.
Pipeline	Structured JSON from Component 2 → headline embedding + body reranking → tiered KB lookup or web scrape fallback → LangChain + Gemma 2B RAG generation → level-conditioned adapted Sinhala text → JASON Output to Component 4
Evaluation	99% context completeness, validated with visually impaired user testing.
Impact	First system to measure and adapt Sinhala text complexity for visually impaired readers

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Adaptive Conversational Personalization and Voice Interaction Module

Interactive Sinhala Text Reader for the Visually Impaired — Component 4

Problem Definition & Novelty

The Challenge

- Existing Sinhala assistive tools convert text to speech passively, with no conversational interaction for the user.
- Current voice interfaces rely on rigid command-based input and cannot recognize user intent or adapt to individual preferences.
- Another limitation is that many systems do not properly support **Sinhala conversational interaction**.



Novelty: User's intent and provides **personalized responses for visually impaired Sinhala users**.

Research Objectives

Main Objective Develop the Adaptive Conversational Personalization and Voice Interaction Module enabling visually impaired users to communicate via natural voice in Sinhala.

01	Develop a Sinhala speech recognition interface converting voice commands to textual input.
02	Design an intent detection mechanism identifying user requests from spoken input.
03	Build a user interaction analysis mechanism recording preferences and interaction patterns.
04	Implement personalized response generation adjusting delivery based on user history.
05	Generate adaptive Sinhala speech responses using TTS for visually impaired users.
06	Evaluate the module using metrics including interaction accuracy, response relevance, and usability.

Research Methodology & Technologies

Research Pipeline

1. **Voice Input Capture**
   User speech is recorded and processed using a Sinhala-compatible STT module.
2. **Intent Detection**
   NLP transformer models classify the user's intent from the transcribed text.
3. **Personalization**
   Adaptive controller stored user preferences and adjusts the request accordingly.
4. **LLM Response Generation**
   Personalized query is sent to the main language model for contextual response.
5. **Adaptive TTS Output**
   Response delivered to the user through as Sinhala TTS output.

Tools & Technologies

Whisper / Sinhala ASR
Speech-to-Text recognition

Python Transformers / BERT
Intent detection & NLP

Scikit-learn / PyTorch
Personalization engine

Sinhala TTS / Google TTS
Adaptive voice output

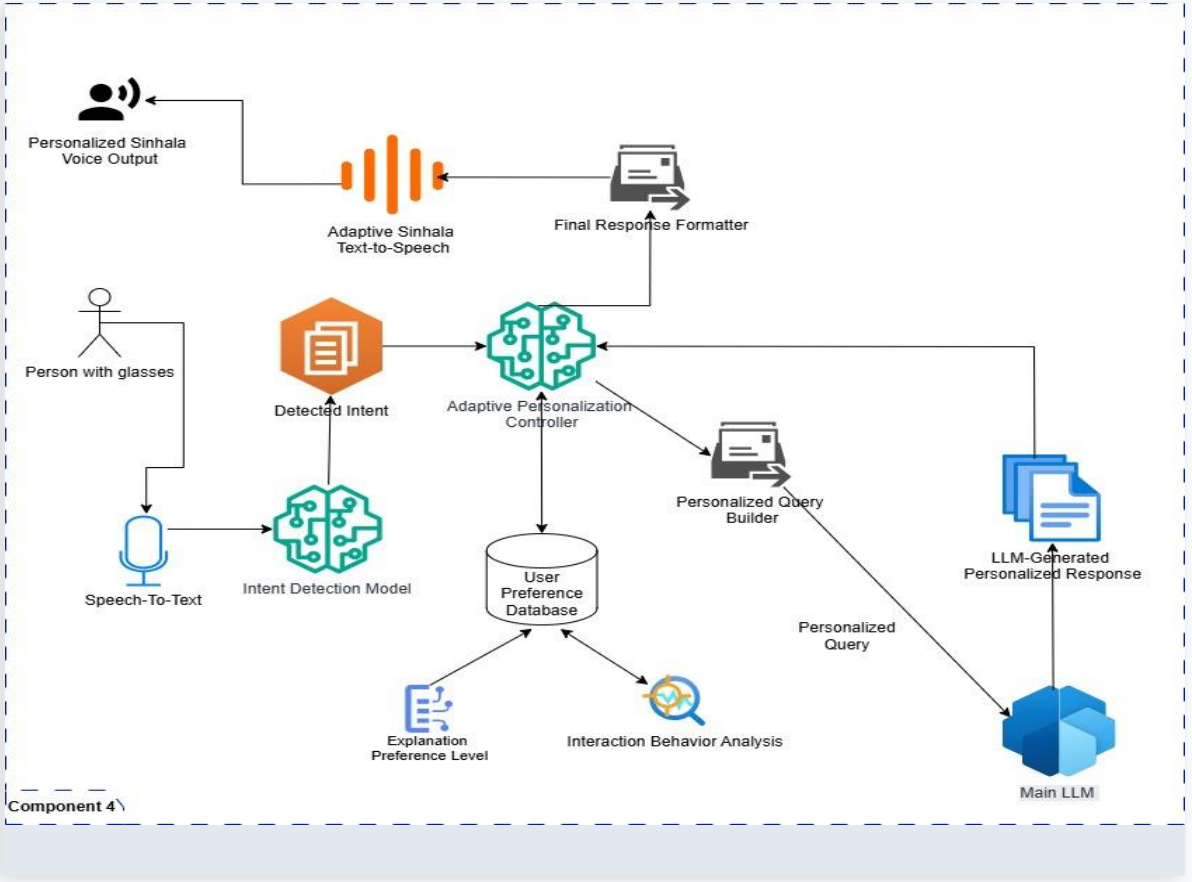
Python Backend
System integration layer

Design Science Research (DSR)
Research methodology

System Overview

Pipeline Stages

- Speech-to-Text (STT) — voice input conversion
- Intent Detection Model — user intent classification
- Adaptive Personalization Controller — preference management and interaction history.
- Personalized Query Builder — LLM query construction
- Final Response Formatter — output structuring
- Adaptive Sinhala TTS — personalized voice delivery



Component 4 receives voice commands, applies personalization, and returns adaptive Sinhala speech via Component 3's LLM.

Dependencies · Validation · Data · Ethics

Dependencies

- Depends on Component 3's LLM for response generation via personalized query input.
- Integrates Sinhala ASR and TTS services for full voice I/O pipeline.
- Requires User Preference Database shared across sessions for personalization.

Validation & Evaluation

- STT performance measured using Word Error Rate (WER) and recognition accuracy.
- Intent detection evaluated with precision, recall, and F1-score metrics.
- User satisfaction and response relevance validated through usability tests with visually impaired participants.

Data Availability

- Open-source Sinhala speech datasets used for ASR training and testing.
- User interaction data collected during system testing with participant consent.
- All personal data anonymized before use in personalization model training.

Ethics & Security

- User data is collected only with informed consent
- User preference data stored securely with appropriate access controls.
- AI models deployed transparently in compliance with institutional research ethics.

Summary

Problem	Sinhala-speaking visually impaired users lack a personalized, conversational voice interface for assistive reading.
Approach	Design Science Research methodology combining STT, intent detection, personalization, and adaptive TTS into one module.
Pipeline	Voice input → STT → Intent Detection → Personalization Controller → LLM query → Response → Adaptive TTS output.
Evaluation	Metrics include WER, F1-score for intent, user satisfaction ratings, and interaction success rate with VI participants.
Impact	Visually impaired users to **access information more easily and interact with documents in a natural way**

Commercialization

Investment and Cost

- Hardware cost is primarily a high-quality auto-focus industrial webcam (LKR 7,000+); all software tools are open-source.
- Google Colab Pro (LKR 9,000) provides sufficient GPU capacity for SinBERT and mT5-small fine-tuning.
- No additional licensing fees; student research time is the primary labour investment.

Pricing Strategy

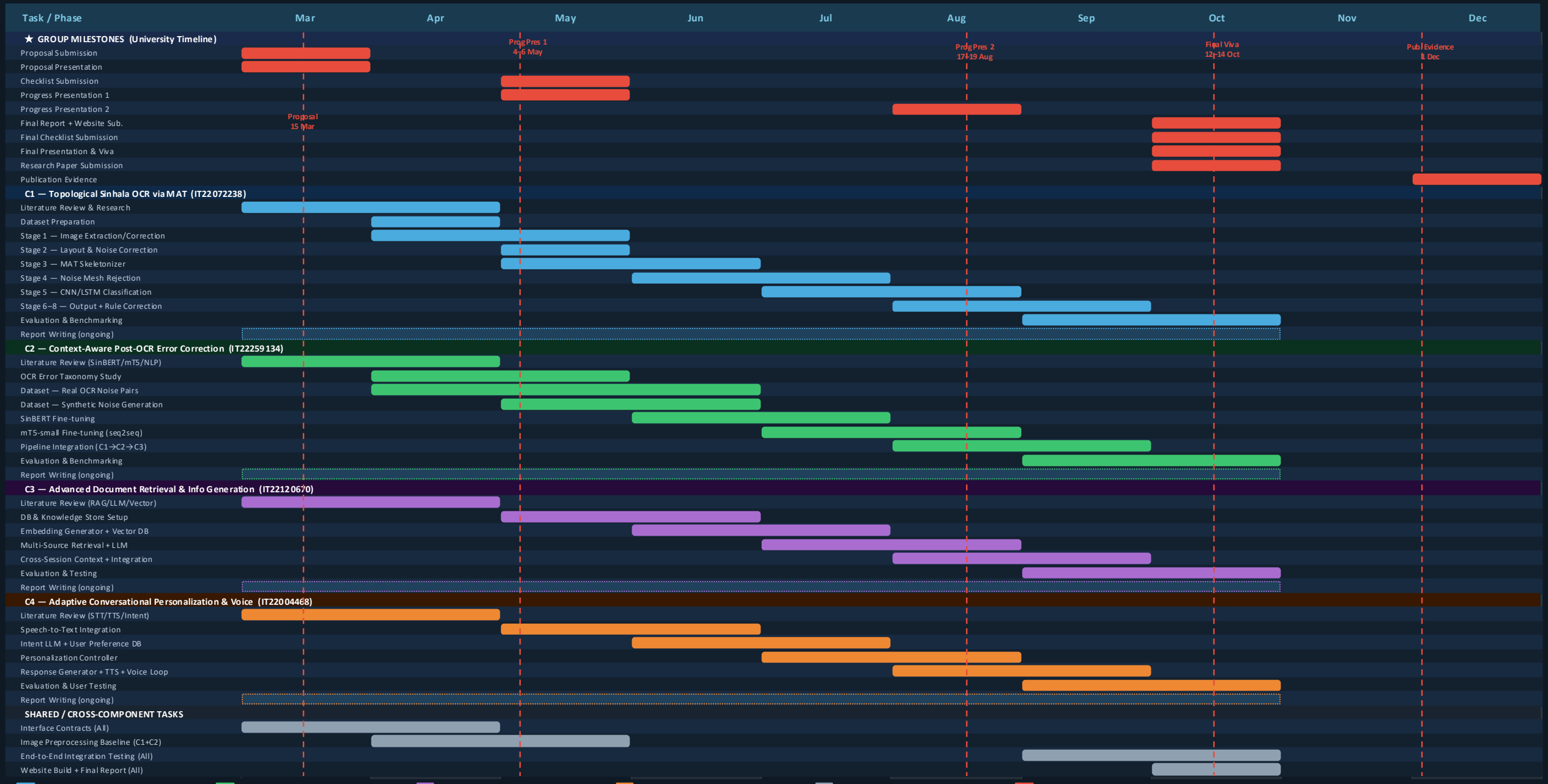
- Component code is released open-source (MIT/Apache 2.0); detailed architecture documentation and commercial rights are offered at a price.
- Full assistive device targets a monthly subscription of LKR 2,500-3,500, significantly below OrCam MyEye at USD 2,000.
- OCR Correction API is licensed to institutions under volume-based pricing for document digitisation workflows.

Target Market and Customer Profile

- Primary segment: 1.1 million visually impaired Sinhala speakers in Sri Lanka who consume newspapers and textbooks.
- Secondary segment: government departments, media publishers, and universities performing Sinhala document digitisation at scale.
- Early adopter persona: medium-sized government department currently using manual proofreaders to fix OCR output.

Value Proposition and IP Rights

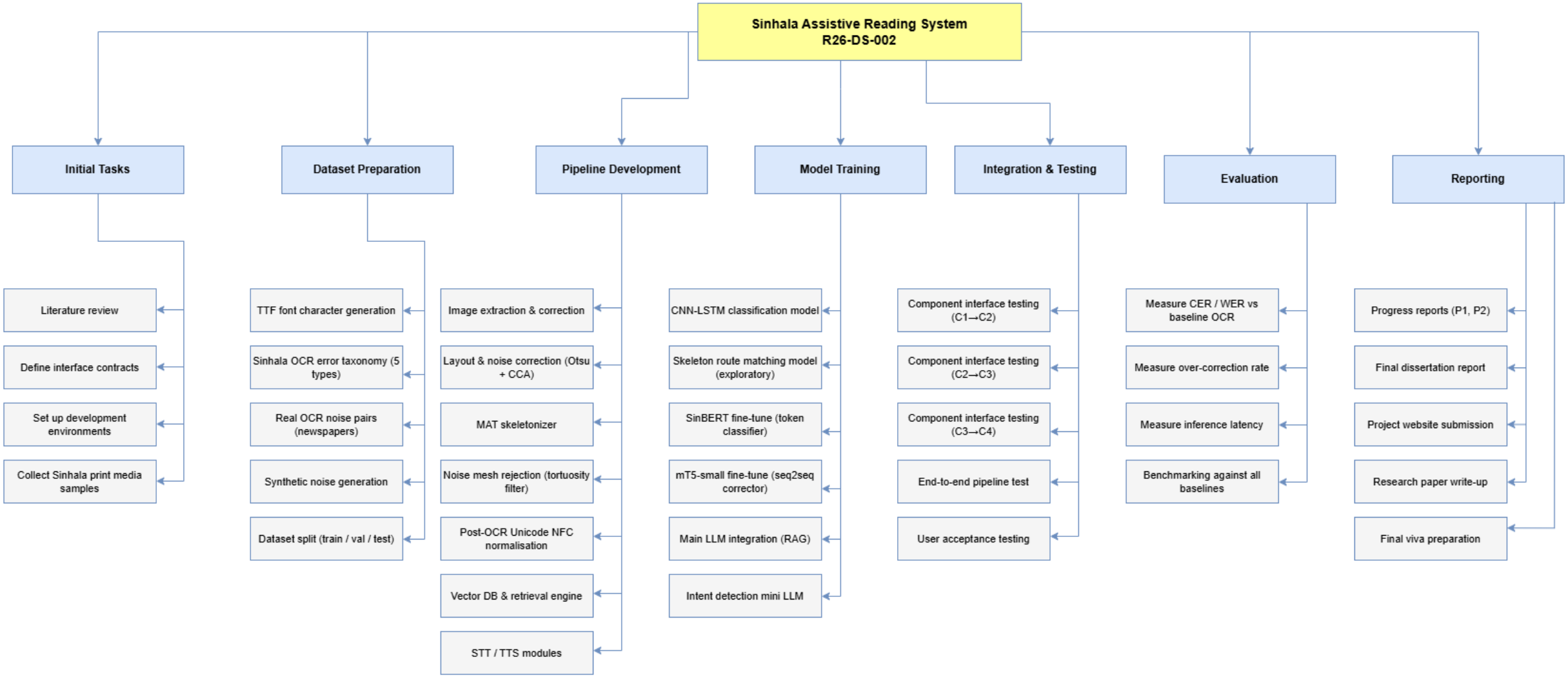
- Unique value: the only context-aware, font-invariant Sinhala OCR correction pipeline designed for real-time offline wearable use.
- Competitive advantage: structural topology approach generalizes to unseen fonts unlike appearance-based models; fully open-source.
- IP protection: released under open-source license with attribution requirements; novel contributions documented in academic publications.



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Interactive Sinhala Text Reader for the Visually Impaired with the assistance of OCR, NLP, and AI.



Thank You!

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